WORKED SOLUTION

STEP I 1997, Question 14

Probability and Statistics • Minimising Expected Cost

STEP I 1997, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q14

The key idea

The cost T is a function of the random variable X, so its expectation is an integral of that function against the density. Splitting the integral at $x = y$ gives $\mathrm{E}(T)$ as an ordinary function of y, and then minimising it is a first-year calculus exercise: differentiate, set to zero, and check that the stationary point really is a minimum.

Solution

We take $y \geq 0$, since a negative amount of preparation makes no sense. Write $T = g(X)$, where

$$g(x) = \begin{cases} y & x \leq y, \\ y + r + s(x - y) & x > y. \end{cases}$$

Then

$$\begin{aligned} \mathrm{E}(T) &= \int_0^\infty g(x)\,\mathrm{e}^{-x}\,\mathrm{d}x \\ &= \int_0^\infty y\,\mathrm{e}^{-x}\,\mathrm{d}x + \int_y^\infty (r + s(x-y))\mathrm{e}^{-x}\,\mathrm{d}x, \end{aligned}$$

where we have pulled out the cost y, which is paid whatever happens. The first integral is y, since the density integrates to 1.

For the second, substitute $u = x - y$, so $\mathrm{e}^{-x} = \mathrm{e}^{-y}\mathrm{e}^{-u}$ and u runs from 0 to ∞:

$$\begin{aligned} \int_y^\infty (r + s(x-y))\mathrm{e}^{-x}\,\mathrm{d}x &= \mathrm{e}^{-y}\int_0^\infty (r + su)\,\mathrm{e}^{-u}\,\mathrm{d}u \\ &= \mathrm{e}^{-y}\left(r\int_0^\infty \mathrm{e}^{-u}\,\mathrm{d}u + s\int_0^\infty u\,\mathrm{e}^{-u}\,\mathrm{d}u\right) \\ &= \mathrm{e}^{-y}(r+s), \end{aligned}$$

using $\int_0^\infty u\,\mathrm{e}^{-u}\,\mathrm{d}u = [-u\mathrm{e}^{-u}]_0^\infty + \int_0^\infty \mathrm{e}^{-u}\,\mathrm{d}u = 1$ (integration by parts). Hence

$$C(y) = \mathrm{E}(T) = y + (r+s)\,\mathrm{e}^{-y}.$$

Ask yourself Is there a way to see the factor $e^{-y}(r+s)$ without integrating?

Yes. Flood damage happens with probability $P(X > y) = e^{-y}$. Given that it happens, the excess $X - y$ is again exponential with mean 1 (the exponential distribution is *memoryless*), so the expected damage is $r + s \cdot 1$. Multiplying gives $(r+s)e^{-y}$.

Now minimise. Differentiating,

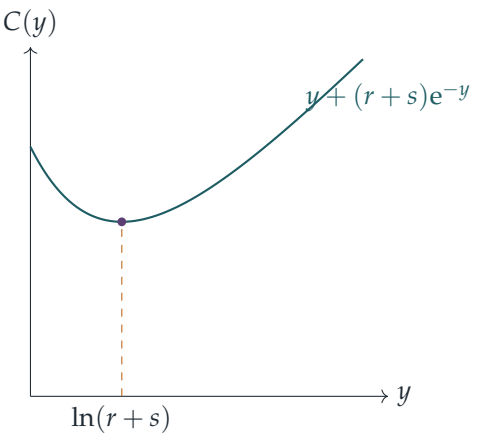
$$C'(y) = 1 - (r+s)e^{-y}, \qquad C''(y) = (r+s)e^{-y} > 0.$$

So $C'(y) = 0$ exactly when $e^y = r+s$, that is $y = \ln(r+s)$. Since $C'' > 0$ for all y, C is convex, C' is increasing, and so C decreases for $y < \ln(r+s)$ and increases for $y > \ln(r+s)$. The stationary point is therefore the global minimum, and the expected total cost is minimised by taking

$$y = \ln(r+s),$$

as required. The minimum expected cost is $C(\ln(r+s)) = \ln(r+s) + 1$.

This needs $\ln(r+s) \geq 0$, i.e. $r+s \geq 1$, which is implicitly assumed. If $r+s < 1$ then $C'(y) > 0$ for every $y \geq 0$, so the best choice is $y = 0$: the damage is so cheap that it isn't worth preparing at all.



(Sketch drawn with $r+s = 3$.)

Checks

- At $y = 0$ we always pay damage, and $E(T) = r + s\,E(X) = r+s$, which matches $C(0)$.
- With $r = 2$, $s = 3$ the formula gives $y = \ln 5 \approx 1.609$ and minimum cost $1 + \ln 5 \approx 2.609$; a fine grid search of $C(y)$ finds the same minimiser, and a simulation of 4×10^5 years at $y = 1$ gave a mean cost of 2.844 against the formula's $1 + 5e^{-1} \approx 2.839$.

- As $r+s$ grows, y increases: dearer floods justify more preparation, as common sense says.

Common slips

- Forgetting that the preparation cost y is paid in *every* year, not only when $X \leq y$.
- Integrating $sx\,\mathrm{e}^{-x}$ but losing the $-sy\,\mathrm{e}^{-x}$ term, which gives $\mathrm{e}^{-y}(r+s+sy)$ instead of $\mathrm{e}^{-y}(r+s)$.
- Finding the stationary point without showing it is a minimum.

Beyond STEP

This is a small example of decision-making under uncertainty: choose an action to minimise expected loss. The same structure, balancing a sure cost against an expected penalty, underlies the "newsvendor problem" in operations research and the idea of Bayes decisions in statistics. The memoryless property used above characterises the exponential distribution among continuous distributions; Grimmett and Welsh, *Probability: An Introduction*, covers it along with expectations of functions of random variables.

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